Smaller Diophantine formulations of Con(ZFC)

Four-cell coefficients, a six-equation degree-28 system,
and single equations in fourteen positive unknowns

Research synthesis prepared for Vladimir Reshetnikov
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Abstract

We combine four compatible reductions from the eight existing ProveIt packages on Diophantine forms of ZFC consistency. The resulting family improves several of their measured bounds. First, the input-free construction and the ordinary strong Pell norm give six equations in fifteen positive unknowns, of maximum degree 28, with a 61-operation evaluator. The previous degree-28 system used sixteen unknowns, seven equations and 63 operations. Its product finalizer has degree 99 and 69 operations; its sum of squares has degree 56 and 78 operations. Second, transferring the existing quadratic-root elimination to the input-free construction gives a single equation in fourteen positive unknowns: either 73 operations and degree 342, or 75 operations and degree 293. Third, the ordinary strong norm removes a register from the existing quadratic lift, giving 24 quadratic equations in 33 positive unknowns and a quartic sum of squares.

A separate compiler reduction replaces nine-cell windows by four directed cells. Moving the predecessor initialization test to its predecessor removes that coordinate as well as all four unused corners. The finite tile alphabet decreases from about $1.42 \cdot 10^{21}$ to 71,356,645. The coefficient $B - 1$ now has exactly 5^{28} binary digits instead of 5^{67}, a reduction by 5^{39}. All coefficients are defined by finite, unconditional recipes; their enormous expansions are not materialized. The report supplies ordinary proofs, explicit arithmetic circuits, the exact machine table, and reproducible standard-library checks. These are improvements within the inspected constructions, not global optimality theorems. The new arithmetic and compiler bridge are not Lean/Rocq formalizations and do not prove consistency.

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1 The result and the comparison being made

Let $\mathrm{Proof}_{\mathrm{ZFC}}(p, \bot)$ be the usual effective finite-refutation predicate, in the presentation calibrated to K215. As usual, Con(ZFC) means $\neg\exists p\ \mathrm{Proof}_{\mathrm{ZFC}}(p, \bot)$. The repository's endpoint identifies this refutation predicate with halting of the literal blank-tape machine K215 [2].

Theorem 1.1 (Fixed consistency representations)**.** *Fix the four integer coefficients of Section 7. Let $\mathcal{S}_{15}$, P_{69}, Φ_{14}, Ψ_{14}, and Q_{33} be the systems and polynomials defined below. ZFC proves that each of the following assertions is equivalent to $\exists p\ \mathrm{Proof}_{\mathrm{ZFC}}(p, \bot)$:*

1. *$\mathcal{S}_{15}$ has a solution in $\mathbb{Z}^{15}_{>0}$;*
2. *$P_{69} = 0$ has a solution in $\mathbb{Z}^{15}_{>0}$;*
3. *$\Phi_{14} = 0$, or equivalently $\Psi_{14} = 0$, has a solution in $\mathbb{Z}^{14}_{>0}$;*
4. *$Q_{33} = 0$ has a solution in $\mathbb{Z}^{33}_{>0}$.*

Consequently, the absence of solutions of each displayed construction is provably equivalent in ZFC to Con(ZFC)*. The explicitly defined nonnegative and unrestricted-integer variants in Section 6 have the same property.*

The proof is the composition of the ordinary K215 calibration, the native Pell/mask compiler, and the changes proved here. The assertion about *provability in ZFC* uses the formalizability of those ordinary arguments. It does not infer an internal ZFC theorem from the soundness of Lean or Rocq. Section 8 states the verification boundary.

1.1 What "simpler" means here

There are different measures: number of unknowns, number of equations, degree, arithmetic operations, coefficient height, and proof complexity. The operation convention follows the existing packages: each binary $+, -, \times$, including multiplication by a fixed numeral, costs one; reading a constant, register reuse, and equality comparison cost zero. Intermediate expressions are substituted polynomial abbreviations, not additional existential variables. A sum-of-squares finalizer is fully charged. Coefficient bit length is reported separately rather than hidden inside the operation figure.

Construction	Positive unknowns	Equations	Degree	Operations
Existing 02 degree-28 system	16	7	28	63
Existing 06/08 basic system	15	6	60	61
New $\mathcal{S}_{15}$	**15**	**6**	**28**	**61**
Existing 06/08 product	15	1	155	67
Existing 06 reduced product	15	1	123	68
New P_{69}	15	1	99	69
Existing 02 sum of squares	16	1	56	83
New sum of squares of $\mathcal{S}_{15}$	15	1	56	78
New Ψ_{14}	14	1	342	73
New Φ_{14}	14	1	293	75
Existing 08 quadratic lift	34	25	2	–
New quadratic lift	33	24	2	–
New Q_{33}, its sum of squares	33	1	4	–

The systems' degree entries are maximum residual degrees; the single polynomials' entries are exact. The original 67-operation polynomial is still the shortest operation count among these single-polynomial evaluators, and can also use the new coefficients. Fourteen is the smallest

positive-variable count among the eight inspected packages and these additions, but it is not a global record: classical nine-unknown representations and the repository's existence theorem are a different comparison class [1, 13]. No minimality assertion is made.

1.2 How the eight packages contribute

The baseline reconciliation is taken from the pinned repository, then the relevant formulas and proof dependencies were read directly. Package 01 gives conservative fixed-input and quadratic alternatives. Package 02 supplies the ordinary strong norm and its integrality/rank proof. Package 03 combines fixed markers with root elimination. Package 04 explores direct two-sided normalization and an affine marker guard. Package 05 provides particularly explicit root-elimination formulas and integer conversions. Packages 06 and 08 independently remove the input marker and reach the same 67-operation arithmetic skeleton; 06 has the smaller, fully numerical raw-K215 coefficient recipe. Package 07 gives alternative scales and degree/operation tradeoffs. Package 08 supplies the 34-variable quadratic lift.

The synthesis is therefore specific: use 06's input-free interface, 02's ordinary norm, 05's root elimination where variable count matters, and 08's quadratic organization. The four-cell coefficient construction is an additional reduction. Coefficients from separate compiler recipes must never be mixed. All four ports in this report come from the single new recipe in Section 7; the complete package-06 tuple remains an independently valid legacy choice for the algebra.

2 The recommended six-equation system

The fifteen unknowns are

$$(J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma) \in \mathbb{Z}_{>0}^{15}.$$

The fixed coefficients are $\beta, K, M_C, M_F^{\mathrm{src}}$. Define

$$\begin{gathered} q = \beta J + 1, \quad C = q - F - \alpha, \quad X = wq, \quad Y = sq^3, \\ E = XY, \quad k = \eta + \zeta, \quad c = kY + \eta, \quad a = E + Y, \\ A = a + 2, \quad H = 4a + 3, \quad \Delta = a^2 + H = A^2 - 1, \\ D = X + ac + \gamma H, \quad G = (q-1)(q-F) + \alpha, \\ R = G(q^2 - 1) + (M_C + qM_F^{\mathrm{src}})J, \quad L_0 = EkY, \\ V = c(Tf - 1) - Rf^2, \quad U = ic^2, \quad W = \Delta(f^2 - 1). \end{gathered} \tag{1}$$

The symbol W is only an auxiliary polynomial in this section; there is no numerical-input marker. The six equations are

$$\boxed{\begin{aligned} \tau^2 - L_0(L_0 + k) &= 1, \\ D^2 - \Delta c^2 &= 1, \\ W(V^2 - y^2) + y^2 &= 1, \\ k - hE - R &= 1, \\ (K + w)C + q - F - z(q-1) &= 1, \\ U^2 &= W. \end{aligned}} \tag{2}$$

Every exponent in these polynomials is a fixed integer. All equalities are ordinary polynomial equalities over the stated domain.

Let N_0, N_1, N_a, N_k, N_t denote the first five left sides and put $N_o = 1 + U^2 - W$. Then a single-polynomial alternative is

$$\boxed{P_{69} = N_0 N_1 N_a N_k N_t N_o - 1 = 0.} \tag{3}$$

Its positive zero set is exactly the solution set of (2), not merely equivalent in nonemptiness. The sum of the six squared residuals is another single equation, of degree 56.

2.1 The normalized parent and the changed proof order

For comparison, the input-free normalized parent has

$$S = (i\Delta c^2)^2, \quad N_a^{\mathrm{n}} = S(V^2 - y^2) + y^2, \quad N_s^{\mathrm{n}} = \Delta f^2 - S,$$

and polynomial

$$P_{67}^{\mathrm{n}} = N_0 N_1 N_a^{\mathrm{n}} N_k N_t N_s^{\mathrm{n}} - \Delta. \tag{4}$$

Its strong equation is $f^2 - \Delta i^2 c^4 = 1$. In contrast, the ordinary equation in (2) is $(ic^2)^2 = \Delta(f^2 - 1)$. These two individual equations are not unconditionally equivalent. The ordinary equation can have extra rational Pell solutions when isolated. It becomes sufficient here because the main Pell ordinate is already large. The central proof change is to establish rank *before* restoring the auxiliary quotient. This is why the combination requires a proof rather than a formal substitution.

3 Proof of the ordinary-norm combination

Write $\chi_A(n) + \psi_A(n)\sqrt{A^2 - 1} = (A + \sqrt{A^2 - 1})^n$. Positive Pell solutions at parameter $A \geq 2$ are classified by these sequences. The two sequences satisfy the same second-order recurrence, and ψ_A has the strong divisibility property.

3.1 Unit signs and the boundary $C = 0$

At a positive zero of (3), every factor is an integer unit. The negative consecutive-product norm $x^2 - u(u+1)v^2 = -1$, for $u \geq 2$, is impossible: a solution minimal in $v > 0$ is reduced by multiplication by $2u + 1 - 2\sqrt{u(u+1)}$, whose new second coordinate is $0 < (2u+1)v - 2|x| < v$. Thus $N_0 = 1$. Since $\Delta \equiv 0$ or $3 \pmod 4$, the main norm cannot equal -1. The ordinary strong factor also cannot be -1: when $\Delta \equiv 0$, it is $1 + U^2 \pmod 4$; when $\Delta \equiv 3$, it is $U^2 + f^2 \pmod 4$. Neither is 3. Hence $N_o = 1$ and $W = U^2$. The auxiliary norm is consequently congruent modulo four to either y^2 or V^2, excluding its negative unit. We have

$$N_0 = N_1 = N_a = N_o = 1, \qquad N_k = \epsilon, \quad N_t = \nu, \qquad \epsilon\nu = 1, \quad \epsilon, \nu \in \{1, -1\}.$$

Transport gives $C \geq 0$, because

$$q - F - z(q-1) = 1 - F - (z-1)(q-1) \leq 0$$

and $C \leq -1$ would make $N_t \leq -(K + w) \leq -2$. One must not conclude $C > 0$ yet: the boundary $C = 0, F = 2, z = 1, \alpha = q - 2$ can make the transport unit negative. The source-mask bounds now give

$$\begin{gathered} 1 \leq F \leq q - 1, \qquad q \leq G \leq q^2 - q, \\ q(q^2 - 1) < R < q^4 - q^3 + q^2 - 1 < q^4 - 2, \\ R - 2 > 3q + 1, \qquad E \geq q^4 > R + 2, \qquad a > R + 2. \end{gathered} \tag{5}$$

Here the shifted mask satisfies $M_F^{\mathrm{src}} < 2(B-1)$, rather than the unshifted bound $M_F^{\mathrm{src}} < B - 1$. These weaker, correct input-free bounds are sufficient throughout.

3.2 Main-index growth independent of the auxiliary sign

Put $P_0 = 2XY^2 + 1$. The first and main norms give positive indices n, p with

$$\tau = \chi_{P_0}(n), \quad k = 2\psi_{P_0}(n), \qquad D = \chi_A(p), \quad c = \psi_A(p).$$

Since $P_0 \equiv 1 \pmod{E}$, the index unit gives $2n \equiv R + \epsilon \pmod{E}$. The positive representative $R + \epsilon$ lies below E, so $n \geq (R-1)/2 \geq 24$. Moreover $P_0 > A$ and $c > k$ force $p > n$. Elementary Pell growth, for example $\psi_A(p) \geq (2A-1)^{p-1}$, gives

$$c > A\Delta^2, \qquad c > 2p, \qquad c > 2R. \tag{6}$$

The last inequality also follows from $k = R + \epsilon + hE$, $E > R + 2$, and $c > kY$. No sign of V was used.

Lemma 3.1 (Ordinary strong rank)**.** *If $c = \psi_A(p) > A\Delta^2$ and $(ic^2)^2 = \Delta(f^2 - 1)$ with positive integers, then some $m > 0$ satisfies*

$$f = \chi_A(m), \qquad p \mid m, \qquad c \mid m, \qquad ic^2 = \Delta\psi_A(m).$$

In particular $c^2 \mid f^2 - 1$.

Proof. This is the ordinary-rank lemma of package 02 and its elementary source proof [4, 7]; we include the argument needed here. Write $\Delta = d_0 s_0^2$ with $d_0 > 1$ squarefree. Let $F_0 + y_0\sqrt{d_0}$ be the least positive integer-coefficient norm-one unit. All positive integer-coefficient units in that ring are its powers, by division by an appropriate power of the least unit. For some $e \geq 1$, with $S_0 = \psi_{F_0}(e)$,

$$A = \chi_{F_0}(e), \quad \Delta = (F_0^2 - 1)S_0^2, \qquad \psi_{F_0}(ep) = S_0 c.$$

The ordinary square equation supplies another unit and an index $v > 0$:

$$f = \chi_{F_0}(v), \qquad ic^2 = T_0\psi_{F_0}(v), \qquad T_0 = (F_0^2 - 1)S_0 \leq \Delta.$$

To justify its integrality, $s_0^2 \mid (ic^2)^2$ implies $s_0 \mid ic^2$; squarefreeness then implies $d_0 \mid ic^2/s_0$, giving the integer second coordinate.

Let $g = \gcd(v, ep)$ and $h_0 = \psi_{F_0}(g)$. Strong divisibility gives

$$c\gcd(c, \Delta) = \gcd(c^2, T_0\psi_{F_0}(ep)) \mid T_0 h_0, \qquad h_0 \geq c/\Delta.$$

If $g < ep$, it is a proper divisor and $2g \leq ep$; Pell duplication gives $2h_0^2 < \psi_{F_0}(2g) \leq S_0 c < Ac$. This contradicts $c > A\Delta^2$ and $h_0 \geq c/\Delta$. Thus $ep \mid v$; write $v = em$. We obtain $p \mid m$, $f = \chi_A(m)$, and $ic^2 = \Delta\psi_A(m)$.

Write $m = pt$. Expansion of $(\chi_A(p) + c\sqrt{\Delta})^t$ gives

$$\frac{\psi_A(pt)}{c} \equiv t\chi_A(p)^{t-1} \pmod{c}.$$

Since $c^2 \mid \Delta\psi_A(m)$ and $\gcd(c, \chi_A(p)) = 1$, we have $c \mid \Delta t$. The Pell expansion modulo Δ gives $\gcd(c, \Delta) = \gcd(p, \Delta)$, which divides p. Consequently $c \mid pt = m$. Finally $p \mid m$ implies $c \mid \psi_A(m)$, and the Pell norm implies $c^2 \mid \Delta\psi_A(m)^2 = f^2 - 1$. □

3.3 Restoring the deleted coordinates

The ordinary equation and (6) imply $f > 2c$ and $f^2 > \Delta + c$. Since $R < a < \Delta$,

$$W - (Rf^2 + c) = (\Delta - R)f^2 - \Delta - c > 0. \tag{7}$$

The auxiliary norm is $WV^2 - (W-1)y^2 = 1$. It excludes $V = 0$. For $v = |V| > 1$, the identity $v^2 - 1 = (W-1)(y^2 - v^2)$ and $y \geq v + 1$ imply $v \geq 2W - 1$. But $T > 0$ and (7) give $V > -(Rf^2 + c) > -W$, excluding negative V of magnitude greater than one. The congruence $V \equiv -c \pmod{f}$, with $f > 2c > 2$, excludes $V = \pm 1$. Thus $V > 1$.

Define, in the proof only,

$$o = cT - Rf, \qquad j = Tf - 1 - R(f^2 - 1)/c.$$

Lemma 3.1 proves that j is integral. The identities $of = V + c$ and $jc = V + R$ prove $o, j > 0$, and $V = of - c = jc - R$. Division has not been introduced into the equation.

The standard two-congruence auxiliary lemma now gives $p = R$. For clarity, it classifies the auxiliary norm at base U: the index is odd, say $\ell = 2v + 1$, and $V = Q_v(U^2)$, where

$$Q_v(1 - A^2) = (-1)^v \psi_A(2v + 1), \qquad Q_v(0) = (-1)^v(2v + 1).$$

Reducing modulo $f = \chi_A(m)$, then using Pell step-down, gives $\ell \equiv \pm p \pmod{m}$. Reducing modulo c, where $c \mid U, m$, gives $R \equiv \pm p \pmod{c}$. The bounds $0 < R, p < c/2$ force $p = R$. This is exactly the inherited interface [4]; all its positivity and integrality premises have now been supplied.

Write $2n = R + \epsilon + vE$, $v \geq 0$. If $v \geq 1$, $E > R + 2$ contradicts $n < p = R$. Hence $2n = R + \epsilon$. The branch $\epsilon = -1$ gives $p = 2n + 1$. Since $2A^2 - 1 > P_0$ and $A > Y + 1$, duplication gives $\psi_A(2n) \geq Ak$, contradicting $c = \psi_A(2n + 1) < k(Y + 1)$. Therefore $\epsilon = \nu = 1$. Only now does transport exclude $C = 0$ and prove $C > 0$.

3.4 Power recovery, masks, and completeness

The retained Pell ratio argument recovers $X = 2^R$; hence $q = B^N$ and $J = (q-1)/(B-1)$. Exact half-binomial extraction gives $\mathrm{pc}(R) \geq 3dN + 2$. Define

$$S_w = (C - 1) + qF, \qquad T_w = (M_C J + 1) + q(M_F^{\mathrm{nat}} J - 1).$$

The source's shifted port gives the exact identity

$$R = (q^2 - S_w)(q^2 - 1) + T_w. \tag{8}$$

Since $0 < S_w < q^2$, the inverse-population lemma forces

$$(C - 1) \,\&\, (M_C J + 1) = 0, \qquad F \,\&\, (M_F^{\mathrm{nat}} J - 1) = 0. \tag{9}$$

The native compiler then decodes a periodic marked computation; the new coefficient proof is supplied in Section 7. It yields a finite blank halting run of K215.

Conversely, a finite halting run gives a normalized positive zero of (4) by the native compiler. Keep every coordinate except replace i_{n} by $i_{\mathrm{o}} = \Delta i_{\mathrm{n}}$. Then $(i_{\mathrm{o}} c^2)^2 = \Delta(f^2 - 1)$ and the auxiliary coefficient is unchanged. This gives a positive solution of (2). Thus the ordinary and normalized systems have the same existence condition, although their individual i-coordinates need not agree. The unit argument proves the asserted exact equality of the ordinary system and product zero sets.

3.5 Exact degrees and operation counts

Write $q_0 = \beta J$, $k_0 = \eta + \zeta$, and $T_0 = w(q_0 - F - \alpha) - zq_0$ for leading forms only. The main norm has the all-value cancellation

$$(X + ac + \gamma H)^2 - (a^2 + H)c^2 = X^2 + 2X\gamma H + (\gamma H)^2 + 2acX + 2ac\gamma H - Hc^2.$$

The six residual degrees are $22, 18, 28, 7, 2, 22$. The product's degree-99 leading form is

$$8q_0^{55} h\gamma i^2 k_0^9 w^7 s^{16} T^2 f^4 T_0. \tag{10}$$

For example the monomial $J^{56} h\gamma i^2 \eta^9 w^7 s^{16} T^2 f^4 z$ has coefficient $-8\beta^{56} \neq 0$. The unique degree-28 residual supplies a nonzero degree-56 square, proving the SOS degree. The emitted source ledgers are

$$\mathcal{S}_{15} : 34M + 27A = 61, \quad P_{69} : 39M + 30A = 69, \quad \text{SOS} : 40M + 38A = 78.$$

The files give every gate and certify that every gate and supplied port is live. Diagnostic line specializations supplement these uniform degree proofs; they do not replace them.

4 Single equations in fourteen positive unknowns

Return to the normalized parent (4) and delete f from its fifteen-variable list. All definitions independent of f are retained. Put

$$\begin{aligned} Q &= (\Delta i c^2)^2, & \mathcal{D} &= 1 + \Delta i^2 c^4, \\ u &= cT, & b_0 &= c + R\mathcal{D}, \\ \mathcal{A} &= Q(\mathcal{D}u^2 + b_0^2 - y^2) + y^2 - 1, & \mathcal{B} &= 2Qub_0, \\ P &= N_0 N_1 N_k N_t, & \mathcal{R} &= \mathcal{A}^2 - \mathcal{D}\mathcal{B}^2. \end{aligned} \tag{11}$$

There are exactly fourteen supplied positive unknowns in these expressions. The two equations are

$$\boxed{\Phi_{14} = P(1 + \mathcal{R}) - 1 = 0}, \qquad 75 \text{ operations}, \quad \deg \Phi_{14} = 293, \tag{12}$$

$$\boxed{\Psi_{14} = [P(\mathcal{A} + 1) - 1]^2 - \mathcal{D}(P\mathcal{B})^2 = 0}, \qquad 73 \text{ operations}, \quad \deg \Psi_{14} = 342. \tag{13}$$

The shorter circuit for Ψ_{14} shares the products in the two squared terms; evaluating the printed display naively need not attain 73. Both complete circuits are included.

4.1 The signed-root issue must be resolved

Algebraic elimination alone restores a nonzero integer f of either sign. The existing signed-quotient theorem was written for the universal compiler and its input bridge. It therefore needs an input-free adaptation before it can be used here.

Lemma 4.1 (Input-free signed quotient)**.** *On either valid coefficient tuple specified in this report, a zero of (4) with every supplied coordinate positive except possibly $T \in \mathbb{Z}$ necessarily has $T > 0$. Equivalently, allowing only f to be signed while $T > 0$ also adds no zeros.*

Proof. The unit exclusions and outer bounds in (5) do not use $T > 0$. Keep $C = 0$ provisionally. Normalized strong divisibility gives $f = \chi_A(m)$, $\psi_A(m) = ic^2$, and in fact $pc \mid m$: first $p \mid m$ by strong divisibility, then expansion of $(\chi_A(p) + c\sqrt{\Delta})^{m/p}$ modulo c gives $c \mid m/p$.

Classify the *absolute* auxiliary root rather than assume its sign. The literal congruences $V \equiv -c \pmod f$ and $V \equiv -R \pmod c$ remain valid. Squaring the first, and combining it with the second and the bounds $0 < p, R < c/2$, gives $p = R$ as in the signed absorption proof [9]. The no-wrap argument needs only $R + 2 < E$, which (5) supplies. Thus $2n = R + 1$, both remaining units are positive, and transport gives $C > 0$. In particular R is odd and $C + F < q$.

Here are the power and population details that prevent a circular sign argument. Put $r = (R-1)/2$ and $\xi = (X+1)^{2r}/X^r$. Pell ratio bounds give

$$Y > \xi/2 - 1 > X^r/3, \qquad a > X^{r+1}/3 > 2^R.$$

The recurrence for $\chi_A(j) - a\psi_A(j)$, whose initial values are 1 and 2, gives $\chi_A(j) - a\psi_A(j) \equiv 2^j \pmod H$. The main-root equation therefore gives $X \equiv 2^R \pmod H$. Both positive numbers are below $a < H$, so $X = 2^R$. It follows that $q = B^N$, with $t = dN$, and $q^3 \mid X$.

The upper ratio estimate, using $r \geq 24$, is

$$0 < c/k - \xi/2 < 16r/(X+1) < 1/2.$$

The exact expansion at $X = 2^{2r+1}$ is

$$\xi = M + \theta, \quad M = \binom{2r}{r} + \sum_{j=1}^{r} \binom{2r}{r+j} X^j, \quad 0 < \theta < 1/4,$$

where M is even. Since $Y < c/k < Y + 1$, we obtain $Y = M/2$. Thus $\nu_2(Y) = \mathrm{pc}(r) - 1$, and $q^3 \mid Y$ implies $\mathrm{pc}(R) = \mathrm{pc}(r) + 1 \geq 3dN + 2$. No sign of V, or congruence $R \equiv 3 \pmod 4$, has entered these deductions.

Apply the word identity (8) and the inverse-population lemma. The masks (9) follow. Since $M_C \equiv 2 \pmod 4$ and $J \equiv 1 \pmod 4$, the low mask forces $C \equiv 1 \pmod 4$. Reducing the source definition of R gives $R \equiv C + M_C J \equiv 3 \pmod 4$.

It remains to use the two unsquared congruences. Let $e = \mathrm{sign}(V)$ and classify the auxiliary Pell norm with odd index $\ell = 2v + 1$. Positive-sign χ step-down gives $\ell = \epsilon R + 2mz$, $\epsilon \in \{1, -1\}$. The two congruences imply, using $c > 2R$ and $f > 2c$,

$$e(-1)^{r+mz} = -1, \qquad e(-1)^{r+(m+1)z} = -1.$$

Dividing proves z even; since $R \equiv 3 \pmod 4$, r is odd and $e = 1$. Consequently $cTf = V + c + Rf^2 > 0$, so $T > 0$. This is the signed parity argument of [10], with the old packed coordinate replaced exactly by C. The input bridge was not used.

Finally, the only direct occurrences of f in the parent are f^2 and Tf. Simultaneously negating f, T preserves every computed value. A negative f with positive T would contradict the first assertion after this sign change. The strong equation excludes $f = 0$. □

4.2 Algebraic deletion and restoration

Before imposing any equation, $\mathcal{D} > c > 0$. Since R is an integer, $b_0 = c + R\mathcal{D} \neq 0$; hence $\mathcal{B} \neq 0$. At a positive zero of (12), the two integer factors are units. Because $\mathcal{B}$ is even, $1 + \mathcal{R} \equiv 1 + \mathcal{A}^2 \pmod 4$ cannot be -1. Therefore $P = 1$ and $\mathcal{R} = 0$. The rational number $f = \mathcal{A}/\mathcal{B}$ satisfies $f^2 = \mathcal{D} \in \mathbb{Z}_{>0}$, so it is an integer: a reduced rational square root of an integer has denominator one. Then $V = uf - b_0$ and

$$QV^2 - (Q-1)y^2 = \mathcal{A} + 1 - \mathcal{B}f = 1, \qquad f^2 - \Delta i^2 c^4 = 1.$$

Thus a signed normalized parent zero has been restored. Lemma 4.1 makes f positive. Conversely, a positive normalized parent zero has $f^2 = \mathcal{D}$, $\mathcal{A} = \mathcal{B}f$, and $P = 1$, so deleting f gives a zero of Φ_{14}. The restored root is unique.

For Ψ_{14}, reduction of (13) modulo P gives $P \mid 1$, hence $P = \pm 1$. Put $f = (\mathcal{A} + 1 - P)/\mathcal{B}$. The equation again implies $f^2 = \mathcal{D}$ and integrality. Its auxiliary norm is $QV^2 - (Q-1)y^2 = P$. As Q is a square, this is a square modulo four and cannot be -1. Thus $P = 1$, and the same restoration applies. The two fourteen-variable polynomials have exactly the same positive zero tuples. An exact all-value identity also holds:

$$\Psi_{14} = P\Phi_{14} + (P-1)(2P\mathcal{A} - 1). \tag{14}$$

The equality of zero sets follows from the unit argument, not from this identity alone.

4.3 Uniform degrees and complete circuits

Deleting the f-dependent cone from the 67-gate parent leaves exactly 50 live producers. The local norm finalizer adds 25 gates; the shared outer norm adds 23. Thus

$$\Phi_{14} : 44M + 31A = 75, \qquad \Psi_{14} : 44M + 29A = 73.$$

The factored computation uses $\mathcal{D}\mathcal{B}^2 = 4(Q\mathcal{D}u^2)(Qb_0^2)$; there is no division gate. With q_0, k_0, T_0 as above, the exact leading forms are

$$\begin{aligned} (\Phi_{14})_{293} &= 8\gamma h i^{12} k_0^{27} w^{21} s^{48} q_0^{177} (q_0 - F)^4 T_0, \\ (\Psi_{14})_{342} &= 64\gamma^2 h^2 i^{12} k_0^{30} w^{26} s^{56} q_0^{206} (q_0 - F)^4 T_0^2. \end{aligned} \tag{15}$$

The respective distinguished monomials and their coefficients are

$$\begin{array}{l|l} J^{182}\gamma h i^{12}\eta^{27} w^{21} s^{48} z & -8\beta^{182} \\ J^{212}\gamma^2 h^2 i^{12}\eta^{30} w^{26} s^{56} z^2 & 64\beta^{212}. \end{array}$$

Both coefficients are uniformly nonzero. An independent sparse-polynomial audit reconstructs these complete highest homogeneous forms exactly.

5 Twenty-four quadratics in thirty-three unknowns

Start with the fifteen unknowns of Section 2, and add the eighteen positive registers

$$N, E, P_*, c, a, X, D, C_2, G, L, R, F_2, V_0, V, g, I, B_a, \Delta.$$

In this section use only the affine abbreviations $q = \beta J + 1$, $k = \eta + \zeta$, $H = 4a + 3$, and $C = q - F - \alpha$. The complete quadratic system is:

1.	$L = q^2 - 1$	2.	$N = q(L+1)$
3.	$X = wq$	4.	$a = E + sN$
5.	$E = X(a - E)$	6.	$c = k(a - E) + \eta$
7.	$P_* = E(c - \eta) + k$	8.	$\tau^2 = P_*(P_* - k) + 1$
9.	$D = X + ac + \gamma H$	10.	$\Delta = a^2 + H$
11.	$C_2 = c^2$	12.	$D^2 - \Delta C_2 = 1$
13.	$k = R + 1 + hE$	14.	$G = (q-1)(q-F) + \alpha$
15.	$R = LG + (M_C + qM_F^{\mathrm{src}})J$		
16.	$(K + w)C + (q - F) - z(q-1) = 1$		
17.	$F_2 = f^2$	18.	$V_0 = Tf - 1$
19.	$V = cV_0 - RF_2$	20.	$g = y^2 - V^2$
21.	$I = iC_2$	22.	$B_a = I^2$
23.	$B_a = \Delta(F_2 - 1)$	24.	$y^2 - B_a g = 1$

Let r_j be left side minus right side. Define the literal polynomial

$$\boxed{Q_{33} = \sum_{j=1}^{24} r_j^2.} \tag{16}$$

The sum has a fixed 24 terms. It contains no hidden variable-length conjunction or computation predicate. The coefficient of τ^4 is 1, so its degree is exactly four.

The projection follows by successive substitution: $N = q^3$, $Y = a - E = sq^3$, $E = XY$, and $P_* = EkY + k$. The six remaining constraints are precisely (2). Conversely every source solution uniquely determines all eighteen registers. The less immediate positivity claims follow from the preceding proof: transport gives $C > 0$, hence $G, R > 0$; the strong-rank argument gives $f > 2c$, hence $V_0 > 0$; auxiliary recovery gives $V > 1$ and $(B_a - 1)g = V^2 - 1 > 0$, hence $g > 0$.

The saving over package 08 is exact. Its scaled-root equation used $I = i\Delta c^2$ and a register $K_2 = D^2 - 1 = \Delta c^2$. After using the ordinary root $I = ic^2$, the register K_2 is needed only by the main norm. Combining its two equations into $D^2 - \Delta C_2 = 1$ removes one register and one equation.

6 Domains: nonnegative and unrestricted integers

All principal equations have been stated over $\mathbb{Z}_{>0}$. Interpreting them unchanged over $\mathbb{Z}$ would be incorrect. For a nonnegative domain, replace every positive unknown x by $n_x + 1$; this preserves the variable count and degree, though a literal evaluator must charge the extra additions.

For unrestricted integers use

$$\mathrm{Pos}(u, v, t) = 1 + u^2 + v^2 + t(t + 1). \tag{17}$$

It is always positive and represents every positive integer. Indeed $4n - 3 \equiv 1 \pmod 4$ is represented by three squares by the three-square theorem; exactly one square is odd. Writing it as $(2t + 1)^2$ and the other two as $(2u)^2, (2v)^2$ gives (17). This also covers $n = 1$.

In both fourteen-variable polynomials, τ, y occur only squared, and neither can vanish at a zero. The first norm excludes $\tau = 0$; the restored auxiliary equation with $Q > 1$ excludes $y = 0$. Thus these two may already range over all integers. Substitute (17) for each of the other twelve coordinates. The resulting single equations have 38 unrestricted integer unknowns:

Integer polynomial	Unknowns	Exact degree	Operations
Φ_{14} after substitution	38	586	159
Ψ_{14} after substitution	38	684	157

Each positivity form costs seven gates. The leading forms in (15) contain neither τ nor y, so the degree doubles exactly. Substitution by disjoint positive quadratic forms preserves a nonzero leading form, as seen by setting two of each triple to zero and obtaining the injective monomial substitution $x \mapsto u_x^2$.

There are also lower-degree integer alternatives. In the quadratic lift, only thirteen original coordinates need positivity; τ, y and all eighteen registers may be signed. The equations determine the registers and the preceding positivity proof makes them positive at every solution. Retain the 33 variables over $\mathbb{Z}$, and introduce three new variables and one equation $x = \mathrm{Pos}(u, v, t)$ for each of those thirteen coordinates. This gives 37 quadratics in 72 unrestricted integers, or one quartic sum of squares. Direct substitution instead removes the thirteen old coordinates and introduces 39, giving 59 integer variables and a degree-eight sum of squares. These improve particular explicit integer lifts in the eight packages; they do not improve the classical universal variable/degree frontier.

7 Smaller fixed coefficients from four directed cells

This section specifies all four numerals used above. The complete finite predicate and exact recipes are also executable in `code/four_cell_compiler.py`; the scalar values are recorded in `data/four_cell_constants.json`. No part of the definition asks whether K215 halts.

7.1 The exact machine and the local alphabet

Use the 430-row table `vendor/k215.tm`, with 215 working states, two symbols, initial state 0 and a two-way blank tape. Its SHA-256 is

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090`.

The halt state is 215; for tableau padding add the two stationary identity actions at that state. The actual first action is $(0,0) \mapsto (1, \mathrm{R}, 93)$.

The cell alphabet has $a_0 = 439$ symbols. Codes 0,1 are headless tape bits. Code $2(q+1)+b$ is bit b with a head in state $q \in \{0, \ldots, 215\}$. Codes 434,435,436 are initial phases L, Q, R, with blank payloads and an initial-state head only at Q. Codes 437,438 are vertical and horizontal boundaries V, H. The old local predicate inspected a nine-cell square but depended only on the five-cell cross. Its preceding-time cell was used solely to detect an H immediately before an initial phase.

Move that test to its predecessor. A new tile is (left, center, right, following). At an H center require a following phase L, Q, R. At an interior center forbid an initial phase on a following interior cell. The horizontal initial grammar remains $L \to L$ or Q, $Q \to R$, $R \to R$, with V forcing L at the left edge and R at the right edge. Vertical boundaries propagate vertically. The horizontal boundary bit propagates across each nonvertical strip; consecutive horizontal boundaries and consecutive vertical boundaries are forbidden.

For an interior-to-interior update, write the current head's prescribed bit, if any, and collect incoming heads from a left neighbor moving right, a stationary center head, or a right neighbor moving left. Two or more incoming heads are forbidden, and the following payload must equal this update. A vertical-boundary neighbor supplies blank headless payload; a horizontal-boundary neighbor is excluded. A head cannot cross a vertical boundary. A following H is allowed only when the center head, if present, is halted; this final boundary step does not perform an interior update. These are the literal tests in the provided predicate.

Lemma 7.1 (Four-cell equivalence and reconstruction)**.** *The old and new local predicates define the same periodic cell planes. For neighboring tiles C, R_t, Y_t, the three equalities*

$$\mathrm{right}(C) = \mathrm{center}(R_t), \qquad \mathrm{center}(C) = \mathrm{left}(R_t), \qquad \mathrm{following}(C) = \mathrm{center}(Y_t)$$

are sufficient to reconstruct such a plane from a cyclic tile word.

Proof. For any interior cell, its predecessor is H exactly when the new rules force that cell to have an initial phase: an interior predecessor forbids such a phase, and a V predecessor forces V rather than an interior cell. All other old clauses are unchanged. This reindexing is valid even if offsets in a cyclic presentation coincide.

For a cyclic word T_i with neighbor offsets $-1, -h_0$, put $u_i = \mathrm{center}(T_i)$. The equalities give the other entries as $u_{i+1}, u_{i-1}, u_{i-h_0}$, respectively. Pull back along $(x,t) \mapsto -x - h_0 t \pmod N$ to obtain a periodic plane. No coprimality assumption is needed. Genuine neighborhoods give the converse. □

The distinguished first tile is $S = (434, 435, 436, 1)$, recording the literal first transition. The distinguished second tile is $E = (437, 437, 0, 437)$. It is valid at its center but has two adjacent

vertical boundaries, so it has no global extension. Enumerate these two first and then all other locally valid four-tuples in lexicographic order. The exact number is

$$\kappa = 71\,356\,645. \tag{18}$$

The transition table has 348 right-moving and 82 left-moving actions. Adding the initial phase gives incoming-head counts 349 and 82 and nonincoming nonhorizontal counts 89 and 356. The center-class census is

Center class	Multiplicity	Valid tiles per center
Vertical boundary	1	192282
Horizontal boundary	1	2634
Headless bit	2	355508
Halt head	2	223966
Working head	430	162788
Initial L	1	1754
Initial Q	1	437
Initial R	1	1750

The accompanying proof and checker refine the working-head row into four direction/write classes and verify every left/right pair for each representative class. The predecessor multiplicity and the unused four-corner multiplicity have both disappeared.

A marked periodic plane still forces a proper finite strip containing the initial L^uQR^v row and exactly one initial head in that strip. Temporal recurrence supplies a later boundary; the last interior head must be halted. Conversely any finite halting run admits independent width and height padding and a helical cyclic presentation. It has one Start and no forbidden tile. A zero temporal stride is impossible since the Start center Q differs from its following center 1. Thus Lemma 7.1 preserves both directions of the blank-machine tableau theorem.

7.2 A finite recipe for every coefficient

The geometry below uses symbols distinct from the Pell unknowns. Let $\ell = 27$, $A_0 = 2^\ell$, and take the ordered slot list

$$\mathcal{L} = ((1,0),(1,1),(1,2),(2,1)).$$

For slot (r,c) and symbol $v < a_0$, put $e(r,c,v) = \kappa + 3a_0 + (2-r)3a_0 + (2-c)a_0 + v$. Define

$$p_0 = 35676757, \qquad j_{\max} = 4a_0 + 4 + p_0 = 35678517,$$
$$\mu = A_0 - 2 + \sum_{j=1}^{j_{\max}} A_0^j, \qquad m_0 = 2\ell + 11a_0 + 8 + 2p_0 = 71358405,$$
$$e_* = \kappa + 12a_0 = 71361913, \qquad M_0 = \kappa + 15a_0 + 1 = 71363231.$$

The native support $\mathcal{E}$ is the disjoint union of selectors $0,\dots,\kappa-1$, the $4a_0$ payload positions, the single dummy position e_*, and anchors $M_0, 3M_0, 9M_0, 27M_0$. It has $m_0 + 1$ elements. For tile T_i, define

$$c_i = 1 + \sum_{j=0}^{3} A_0^{1+ja_0+T_i[j]} + \sum_{t=1}^{4} A_0^{4a_0+t},$$
$$c_{e(\mathcal{L}_j,v)} = A_0^{1+ja_0+v}, \qquad c_{3^{t-1}M_0} = A_0^{4a_0+t}, \qquad c_{e_*} = 0.$$

The remaining fixed integers are

$$\begin{aligned} E_{\max} &= 27M_0 = 1926807237, & H_C &= 51M_0 + 3a_0 + 1 = 3639526099,\\ T_1 &= H_C + 2E_{\max} + a_0 + 1 = 7493141013, & T_2 &= T_1 + 4E_{\max} + 2 = 15200369963,\\ L_C &= 5^{15} = 30517578125, & b &= 5^{13} = 1220703125,\\ \rho &= 2^b, & d &= bL_C = 5^{28}, \qquad B = \rho^{L_C} = 2^d. \end{aligned}$$

All sums below are over fixed finite sets. Define

$$\begin{aligned} D_C &= \rho^{3a_0} + \rho^{H_C+a_0} + \rho^{8M_0} + \rho^{24M_0} + \sum_{e\in\mathcal{E}} c_e(\rho^{T_1-e} + \rho^{T_2-e}),\\ D_R &= \rho^{H_C},\\ M_F^0 &= \mu(\rho^{T_1} + \rho^{T_2}) + \sum_{v<a_0} \rho^{e(1,1,v)}\\ &\qquad + \sum_{v<a_0} \left(\rho^{H_C+e(1,0,v)} + \rho^{H_C+e(1,1,v)}\right) + \rho^{9M_0} + \rho^{27M_0},\\ M_C &= B - 1 - \sum_{\substack{e\in\mathcal{E}\\ e\neq 1}} \rho^e - 2\rho^{e_*}, \qquad M_F^{\mathrm{nat}} = M_F^0 + 4. \end{aligned} \tag{19}$$

The four coefficients of every principal construction are exactly

$$\boxed{\beta = B - 1, \qquad K = D_C + BD_R, \qquad M_C, \qquad M_F^{\mathrm{src}} = M_F^{\mathrm{nat}} + B - 1.} \tag{20}$$

The transition table and finite lexicographic enumeration make these specific integers. No free program index, unspecified existential coefficient, or oracle occurs in (20).

7.3 Why the smaller layout satisfies the compiler hypotheses

The selector clause uses only $\kappa < A_0$, a common low selector coefficient 1, and disjoint powers of A_0 for payloads and anchors. Its proof is independent of whether there are nine or four payload positions. The vertical target $e(1,1,v)$ receives the following payload through shift $3a_0$; the temporal word supplies its center. The horizontal targets at $H_C + e(1,0,v)$ and $H_C + e(1,1,v)$ give the two horizontal equalities. The anchor targets remain $9M_0, 27M_0$. These are precisely the three copy conditions of Lemma 7.1 and the original alignment conditions.

All nonanchor native support lies below M_0. The copy bands and the two center bands are separated by the displayed inequalities

$$T_2 - T_1 > 2E_{\max}, \quad L_C - (T_2 - T_1) > E_{\max}, \quad T_2 + E_{\max} < L_C.$$

There are no accidental contributions to the listed targets. The anchor difference $18M_0$ occurs in the native support in the unique way $27M_0 - 9M_0$; it still forces the temporal rotation to align to a whole cell. An occupied cell cannot agree with an empty neighbor at its one-hot copy coordinates, so occupancy propagates from Start around the entire cyclic word. No diagonal copy condition is needed.

The fixed value $b = 5^{13}$ exceeds the sufficient radix threshold 963319989. In particular

$$\rho \geq \max\{4(m_0 + 3)(2\sum_e c_e + 6) + 8,\ 2\mu + 4,\ 16\}.$$

This is the same conservative no-carry bound as in the parent compiler. The extra dummy bit's spill stays below M_0; at least one of the two center bands is clean under any initial binary rotation. The two anchor parity tests then exclude every nonzero within-radix rotation, and

their unique difference excludes every nonzero within-cell rotation. The inherited clean-band and anchor proof therefore applies literally to the new support [3, 8].

Population accounting changes from twelve to three copy blocks:

$$\mathrm{pc}(\mu) = \ell + 4a_0 + 3 + p_0, \quad \mathrm{pc}(M_F^0) = 2\,\mathrm{pc}(\mu) + 3a_0 + 2 = m_0.$$

Consequently

$$\begin{aligned} 0 < M_C, M_F^{\mathrm{nat}} < B - 1, \quad M_C \equiv 2 \pmod 4, \quad M_F^{\mathrm{nat}} \equiv 4 \pmod 8, \\ \mathrm{pc}(M_C) = d - m_0 - 1, \quad \mathrm{pc}(M_F^{\mathrm{nat}}) = m_0 + 1, \quad \mathrm{pc}(M_C) + \mathrm{pc}(M_F^{\mathrm{nat}}) = d. \end{aligned} \tag{21}$$

These are the exact mask conditions used in the arithmetic and signed proofs. Since $T_2 - T_1 \equiv 2 \pmod 4$, $a_0 \equiv 3 \pmod 4$ and $\rho \equiv 2 \pmod 5$, the paired center terms cancel modulo five and

$$2D_C - D_R \equiv 3 \pmod 5. \tag{22}$$

Thus no conditional correction coefficient is needed.

Completeness must tune the actual packed index, rather than choose an independent exponent. Take independently padded powers-of-five width and height with $N > 25d$, and temporal stride h_0. Changing a low dummy bit in cell j changes the index by $-\Gamma B^j$, where

$$\Gamma = \rho^{e_*}(q^2 - 1)[1 + q(D_C + BD_R + B^{h_0})]$$

is invertible modulo dN by (22). The inherited Boolean subset lemma tunes $R \equiv dh_0 \pmod{dN}$ without altering any tile or copy clause. The canonical no-carry margin still gives $2C + F < q/3$, so $\alpha = q - F - C > 0$. The normalized Pell converse supplies all remaining positive witnesses. This establishes the new coefficient compiler in both directions.

7.4 What the height improvement does and does not say

The old raw-K215 coefficient $B - 1$ had 5^{67} binary digits. The new coefficient has 5^{28}, so its bit length is smaller by the exact factor

$$5^{39} = 1\,818\,989\,403\,545\,856\,475\,830\,078\,125.$$

The exact new bit lengths are

Coefficient	Binary digits
β and M_C	37 252 902 984 619 140 625
M_F^{src}	37 252 902 984 619 140 626
K	41 695 683 867 187 500 001

For K, use $D_C < B$ and $K = 2^{b(L_C+H_C)} + D_C$. The coefficients remain far too large to expand in the archive. This is an exact finite specification with a substantial height reduction, not a claim of small decimal coefficients or an efficiently materializable polynomial.

8 Logical and executable verification status

The source snapshot is pinned on the title page. The repository records closed Lean and Rocq endpoint checks for the literal K215 table; the Lean statement is `KEndpoint215.blank_halts_iff_zfc_refutation`. We inspected the statement and the repository's verification record. Those proof-assistant builds were not rerun. The theorem refers to the effective predicate

`BProv`; ordinary effective proof translations connect it to the chosen conventional formulation of Con(ZFC).

The compiler and all improvements in this report have ordinary mathematical proofs, not new kernel-checkable proofs. Their arguments use finite strings and tables, integer arithmetic, Pell recurrences, finite combinatorics, and standard number theory. The K215 calibration is supplied by the repository's ordinary proof which the formal endpoint mirrors. These arguments can be carried out in ZFC. No appeal to "Lean is sound, therefore ZFC proves the statement" is made.

The included verifiers do establish concrete, reproducible facts:

- exact binding of all 430 transition rows and the dense-table hash;
- the four-cell census, scalar layout and mask populations;
- local/copy checks on 32 padded halting-control tableaux, rejection of a truncated nonhalting prefix, and 10,566 sparse toy support checks;
- gate topology, liveness, complete operation ledgers, and literal agreement of the arithmetic packets with their formulas;
- exact all-ring root-elimination identities and exact independent sparse proofs of the degree-293 and degree-342 leading forms;
- all 24 quadratic residuals, their symbolic projection identities, the register counts, and the stated integer-domain conversions.

They do not decide whether K215 halts, materialize the four giant coefficients, or prove the compiler theorem by finite testing. The small coefficient substitutions used in arithmetic diagnostics are not claimed to encode K215 or any accepting computation. Halting-control machines in the local tableau checks are deliberately modified test machines.

If ZFC is consistent, the equivalences do not allow it to prove these equations have no solutions: that would give its own consistency. Mere consistency alone is not asserted to give the stronger two-sided unprovability of both solvability and unsolvability. The result is a representation of the consistency assertion, not a consistency proof.

9 Further research questions

1. **Formalize a reusable signed quotient theorem.** The delicate steps are the initially possible $C = 0$ boundary, power recovery before parity, and use of both unsquared auxiliary congruences. A theorem stated at this interface would cover the input-free and universal cases.
2. **Formalize the ordinary strong-rank interface.** Its conclusion $p \mid m$, not only $c \mid m$, is what restores the deleted quotient integrally. Packaging both conclusions prevents a recurring proof-order gap.
3. **Compress the local alphabet further.** Mark ordinary cells as left or right of the unique head and enforce a local $V\,L^*\,\mathrm{Head}\,R^*\,V$ grammar. This may reduce legal neighborhoods to a much smaller family at the price of a few extra symbols. No such compiler is claimed here.
4. **Use the C222 proof calculus directly.** A direct local calculus or stack encoding might reduce coefficient height much more than shaving states from the already small machine. It needs a complete fixed-arity encoding and a calibrated consistency equivalence.

5. **Reduce fourteen unknowns without a degree explosion.** The present elimination removes one root but raises degree. Can an ordinary strong-norm elimination preserve its lower degree while controlling rational-root denominators and signs?
6. **Reduce the 33-register lift.** Search for another quadratic relation allowing an intermediate register to be removed without adding an inequality or silently imposing positivity.
7. **Make a materializable coefficient benchmark.** Operations, degree, and variable count omit the dominant storage cost. Compare constructions with explicit bounds on coefficient bit length and on the time needed to produce their expanded coefficients.
8. **Extract a concrete nine-unknown specialization.** The abstract existence result is already known; an explicit specialization with a fully bound index and a reproducible coefficient description would allow an honest comparison with the constructions here.

A Files, reproduction, and provenance

The archive is a self-contained integration directory. It includes the LaTeX source and compiled PDF; the exact K215 table; the original 67-operation circuit as immutable JSON data; every newly emitted circuit; the expanded quadratic systems; detailed proof notes and independent reviews; selected unchanged upstream reference notes; and standard-library Python verifiers. The scripts read the vendored parent data rather than executing inherited repository programs. The directory is self-contained for these finite checks and the PDF build, but does not include the full upstream Lean/Rocq development or every historical proof dependency.

```
python3 verify.py
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

Individual builders compare the committed data by default. Their `--write` options regenerate only their own packets. All new fixed coefficients are the tuple in `data/four_cell_constants.json`. The manifest records file hashes and the source commit. Supplied license and attribution notices preserve the original machine contributors.

The ordinary system and quadratic variants build on packages 02, 06, and 08. The fourteen-variable construction builds on package 06, package 05, and the repository's complete90/92 norm constructions. The four-cell recipe modifies package 06's local compiler while preserving its coefficient conventions and arithmetic interface. The improvements are not attributed to the inherited source authors; their prior contributions are identified separately throughout.

References

[1] ProveIt, *Con(ZFC) in Diophantine form*, reconciliation of eight packages, pinned snapshot above. https://github.com/VladimirReshetnikov/ProveIt/tree/3b201cdea32d2c816f5f3d20734051f25fd3ef14/SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/con-zfc-diophantine-forms.

[2] ProveIt, *Formal correctness of the exact 215-state machine*; and `Computability/BusyBeaver/Lean/BusyBeaver/ZFC/KEndpoint215.lean`. https://github.com/VladimirReshetnikov/ProveIt/tree/3b201cdea32d2c816f5f3d20734051f25fd3ef14/Computability/BusyBeaver/Research/ZFC/k225-formal/verification/k215.

[3] Package 06, *Explicit Diophantine equations for Con(ZFC)*, `06-blank67/report.tex`, and `06-blank67/proofs/fixed_blank_67_proof.md`, in [1].

[4] Package 02, `PELL_INTERFACE_LEMMAS.md` and `FIXED_INPUT75_ORDINARY_PROOF.md`, in [1].

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[7] ProveIt, *Recovering integrality after relaxing an auxiliary Pell norm*, `Computability/HilbertTenthProblem/Papers/1980/PELL_RELAXED_AUXILIARY_PROOF.md`.

[8] ProveIt, native stream/queue compiler research notes, especially `complete75_half_binomial_compiler.md` and the asymmetric-scale and normalized strong-output proofs. https://github.com/VladimirReshetnikov/ProveIt/tree/3b201cdea32d2c816f5f3d20734051f25fd3ef14/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue.

[9] ProveIt, `complete84_signed_quotient_absorption.md`, in the directory of [8].

[10] ProveIt, `complete84_signed_quotient_soundness_scout.md` and `review_complete84_signed_quotient_soundness.md`, in [8].

[11] ProveIt, `complete90_signed_root_elimination.md`, in [8].

[12] ProveIt, `complete92_norm_unit_guard.md`, in [8].

[13] ProveIt, `Jones1982.nine_unknowns`, `Computability/HilbertTenthProblem/Lean/Diophantine/Paper1982/NineMain.lean`. The cited repository theorem is an existence result; no concrete nine-variable equation is emitted by this report.